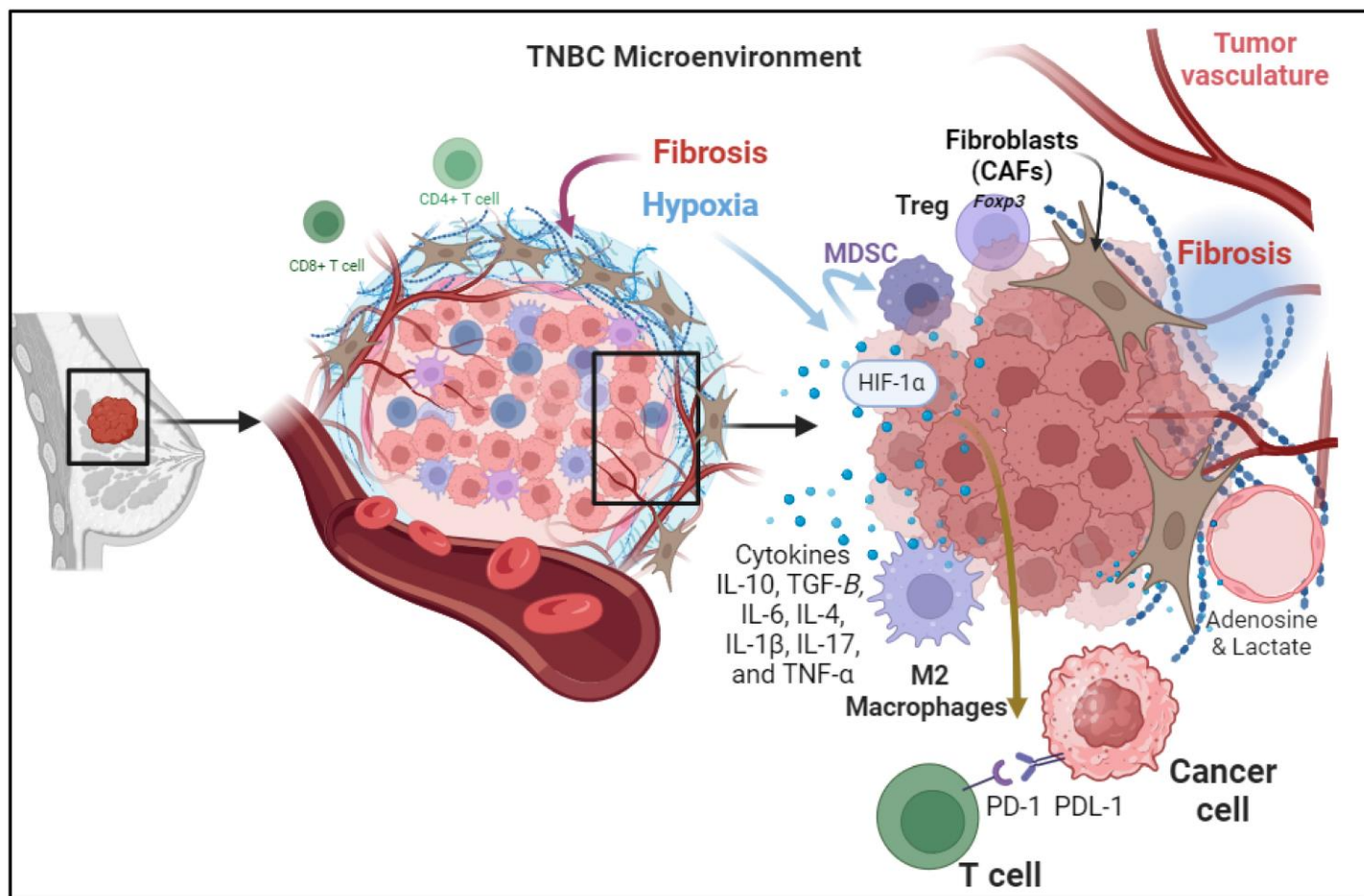
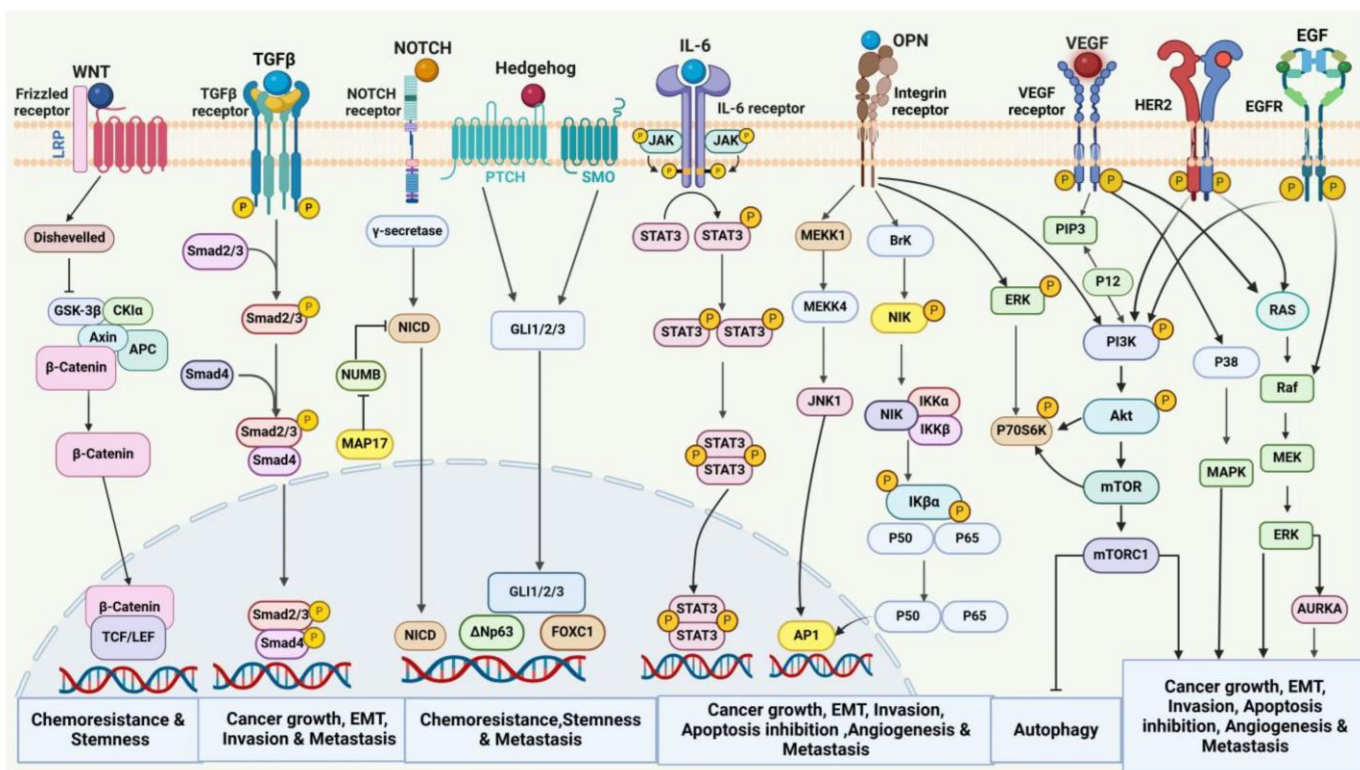


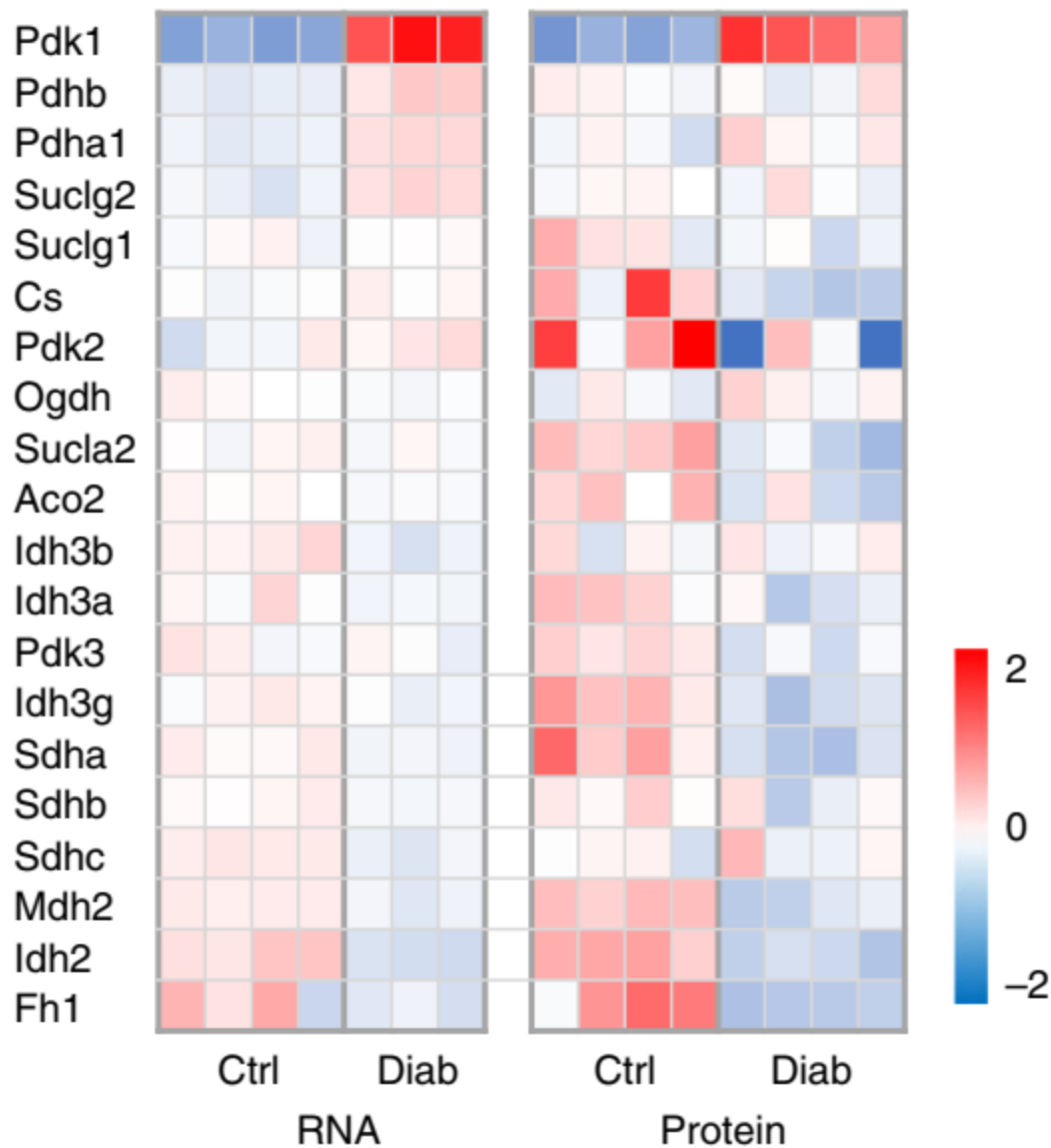
CL-MODE Graphical Abstract

Top Panel — TNBC Problem & Multi-Modal Inputs

A clean horizontal strip showing the biological challenge and the multi-omic inputs.







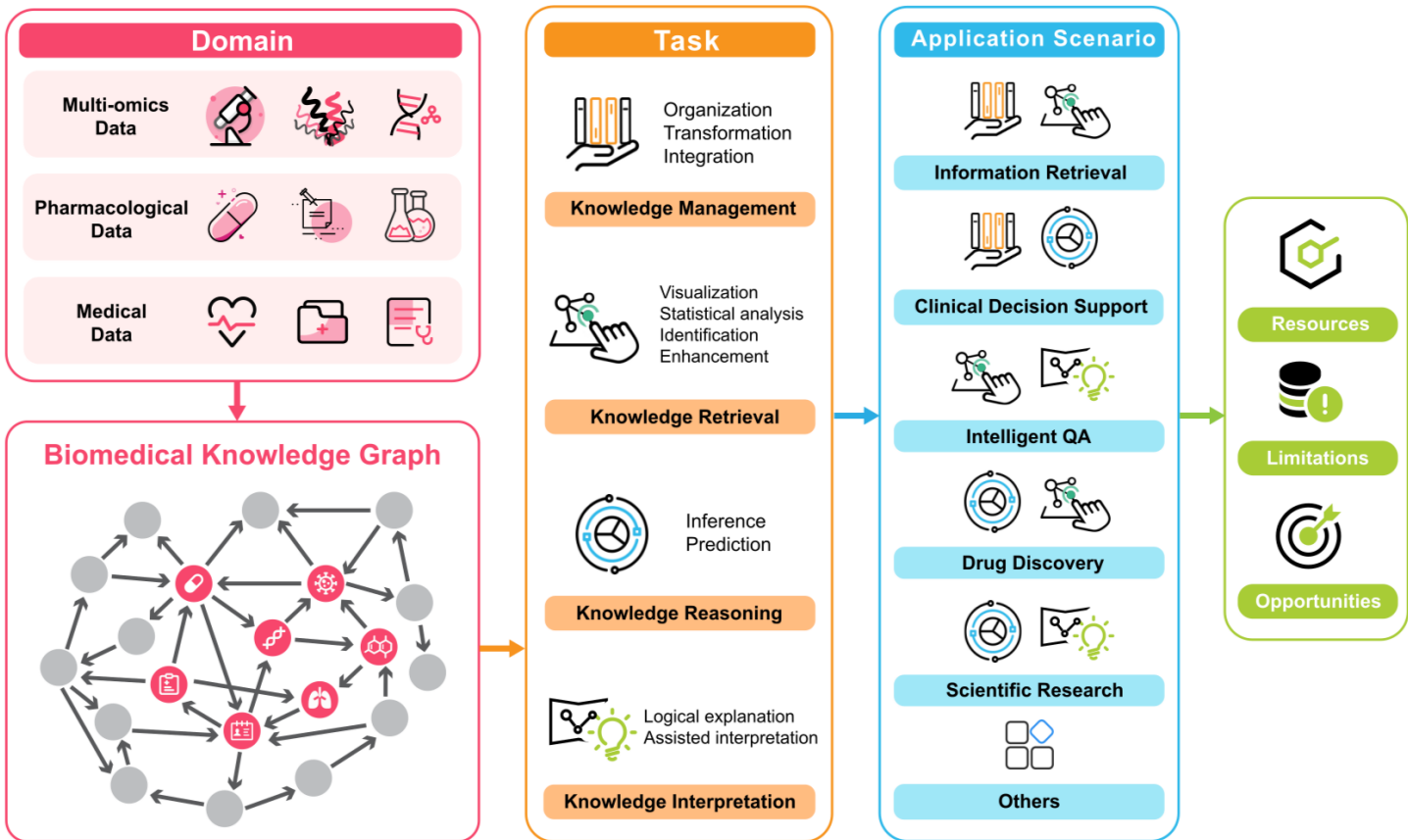
Concept: TNBC is heterogeneous and driven by redundant signaling (PI3K/AKT, MAPK, JAK/STAT, DNA-damage). CL-MODE begins by ingesting **multi-modal omics**:

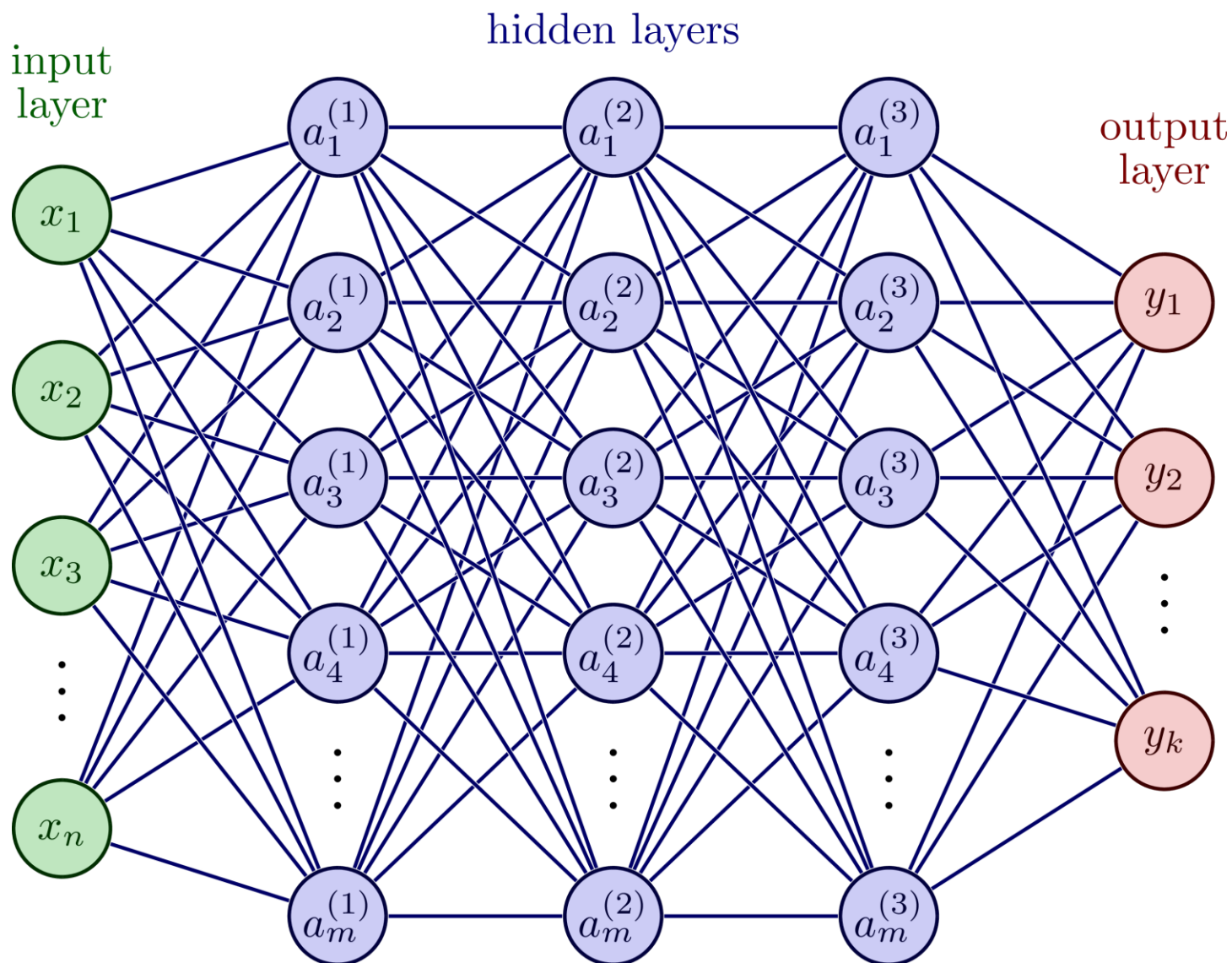
- Genomics (mutations, CNVs)
- Transcriptomics (RNA-seq, pathway scores)
- Proteomics & phosphoproteomics (signaling activation)
- Clinical context (TNBC subtype, immune infiltration)

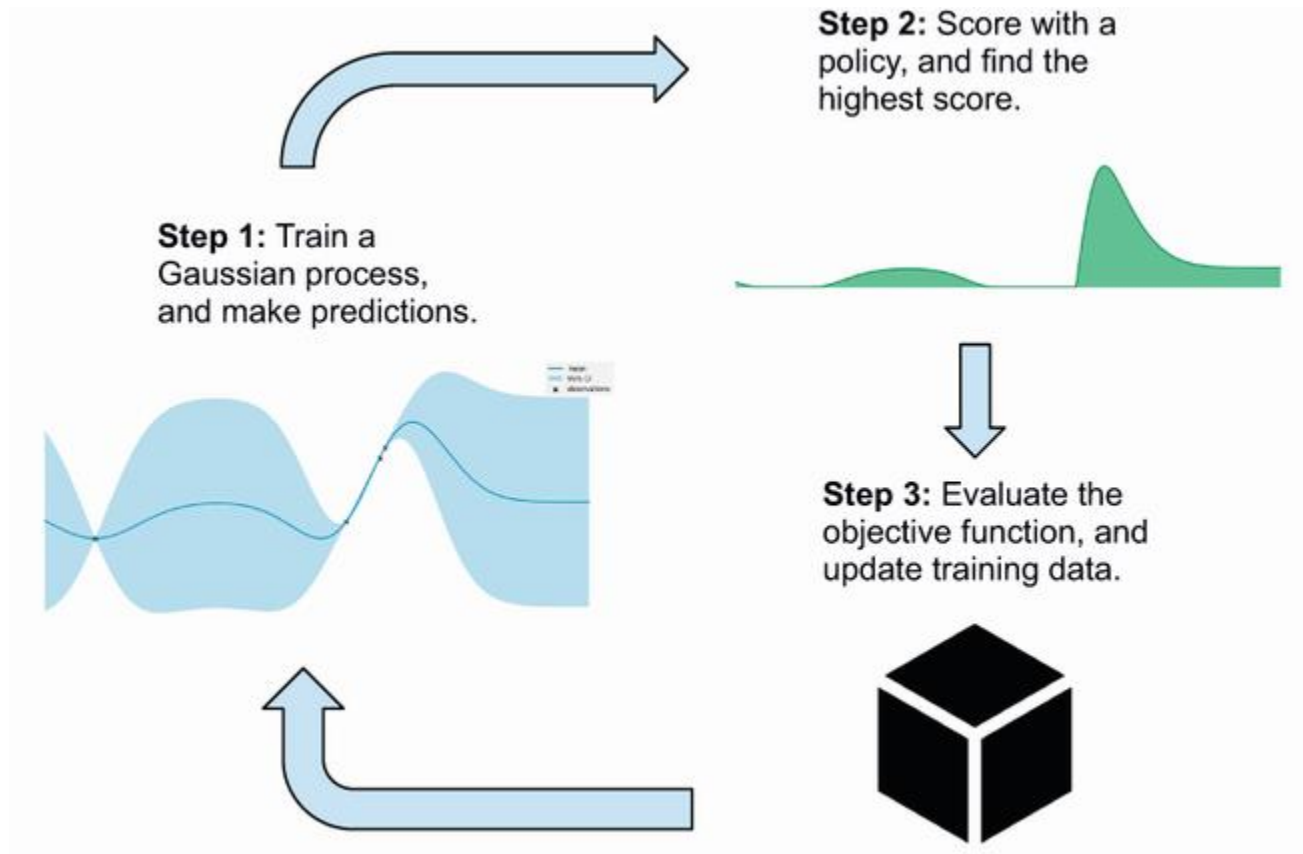
Output: A **tumor state vector** summarizing the functional signaling landscape.

Middle Panel — Core CL-MODE Engine

This is the centerpiece: a three-block pipeline with arrows showing the closed loop.







1. Mechanistic Biomedical Knowledge Graph (KG++)

- Nodes: drugs, targets, pathways, phosphosites, cell lines, trials
- Edges: inhibits, activates, synergizes-with, mutated-in, causes-AE
- Provides mechanistic grounding and interpretability

2. Multi-Modal GNN Synergy Engine (MM-SynergyNet)

- Inputs: tumor state vector + KG++ + drug molecular graphs
- Outputs:
 - Synergy score
 - Selectivity score
 - Mechanistic explanation subgraph
- Highlights RTK–NRTK bypass circuits (EGFR→SRC, MET→ABL1, FGFR→JAK2)

3. Multi-Agent Oncology Reasoning Collective

Agents:

- Genomics

- Proteogenomics
- Synergy
- Mechanistic KG
- Clinical Evidence (RAG)
- Toxicity/Safety
- Chair/Consensus

Agents debate synergy plausibility, toxicity, trial evidence, and resistance durability.

4. Adaptive Experimentation Loop

- Bayesian optimization
- Real or simulated assays
- Feedback updates GNN, KG++, and agent priors
- Reinforcement learning optimizes exploration vs. exploitation

Bottom Panel — Output: Optimal TNBC Drug Combinations

A clean ranked list with mechanistic icons.

DRUG INTERACTIONS



**DRUG - DRUG
INTERACTION**



**DRUG - FOOD
INTERACTION**



**DRUG - HERB
INTERACTION**



**DRUG - DISEASE
INTERACTION**



**DRUG - ALLERGY
INTERACTION**



**DRUG - GENE
INTERACTION**

High-Synergy Pairs

- EGFR + MET
- SRC + ABL1
- FGFR1 + JAK2
- mTOR + DNA-damage agents
- HDAC + RTK inhibitors

High-Synergy Triplets

- **EGFR + MET + SRC**
- **FGFR1 + JAK2 + mTOR**
- **EGFR + HDAC + DNA-damage**

Outputs Provided

- Ranked combinations
- Mechanistic explanation subgraphs
- Tumor-selectivity scores
- Safety flags
- Resistance-durability assessment

Graphical Abstract (Text-Only Layout for Illustrator)

TITLE: CL-MODE — Closed-Loop Multi-Modal Oncology Design Engine

TOP STRIP:

TNBC heterogeneity → Multi-modal inputs:

- Genomics (mutations, CNVs)
- Transcriptomics (RNA-seq)
- Proteomics / phosphoproteomics
- TNBC subtype & immune context

Output: Tumor State Vector

CENTER PANEL (3 blocks + loop):

Block 1: Biomedical Knowledge Graph (KG++)

- Drugs, targets, pathways, phosphosites, trials
- Mechanistic edges (inhibits, activates, synergizes-with)

Block 2: Multi-Modal GNN Synergy Engine (MM-SynergyNet)

- Inputs: Tumor State Vector + KG++ + drug graphs
- Outputs: Synergy, Selectivity, Mechanistic Subgraph

Block 3: Multi-Agent Oncology Reasoning Collective

- Genomics, Proteogenomics, Synergy, Mechanistic KG,
Clinical Evidence, Safety, Chair

Closed Loop:

- Adaptive Experimentation (Bayesian optimization)
- Real/simulated assays → feedback → update GNN + KG++ + agents

BOTTOM STRIP:

Output: Optimal TNBC Drug Combinations

- EGFR + MET
- SRC + ABL1
- FGFR1 + JAK2
- mTOR + DNA-damage
- HDAC + RTK

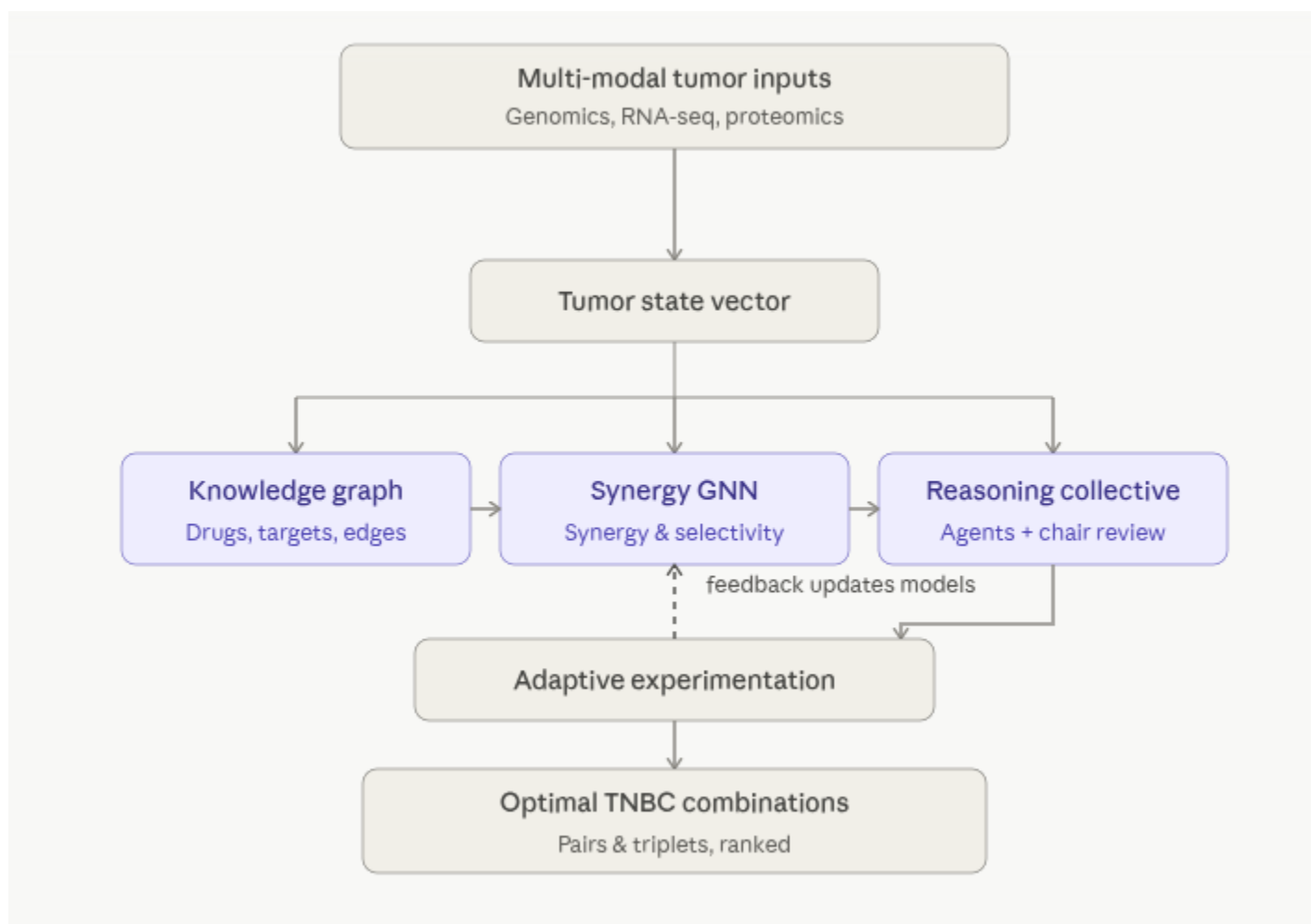
Triplets:

- EGFR + MET + SRC
- FGFR1 + JAK2 + mTOR
- EGFR + HDAC + DNA-damage

With:

- Mechanistic explanations
- Selectivity scores

- Safety flags
- Resistance durability



That's the full architecture: inputs feed a tumor state vector, which drives the three engine blocks (knowledge graph, synergy GNN, reasoning collective) running left to right, with the dashed line showing the adaptive-experimentation feedback closing the loop back into the synergy GNN, ending in ranked combination outputs.

A couple of things worth flagging about the specific candidates in your bottom strip, since they matter for whether this is a good hypothesis set or not:

- **EGFR + MET** and **FGFR1 + JAK2** are reasonable choices for basal-like TNBC — both pairs target known bypass-resistance axes (MET amplification is a classic escape route from EGFR inhibition), so the mechanistic rationale writes itself for the Hypothesis Generator agent.
- **SRC + ABL1** is more of a "generally essential kinase pair" risk — this is exactly the kind of combination that scores well on cell-autonomous cytotoxicity but poorly on tumor selectivity, so it's a good test case for whether your selectivity head is actually working.

- **mTOR + DNA-damage** and **HDAC + RTK** both have real literature support in basal-like/mesenchymal TNBC subtypes, particularly around replication stress sensitization.
- The triplets are where your therapeutic-index modeling earns its keep — three-way combinations compound toxicity risk faster than they compound efficacy, so these are the ones where the Pareto-frontier reporting (Section 5 of the design doc) matters most, rather than a single blended score.

Input from Hypothesis Generator:

Proposed triplet: EGFR inhibitor + MET inhibitor + SRC inhibitor, for basal-like TNBC with EGFR amplification. Rationale: MET amplification is a documented bypass-resistance mechanism when EGFR is inhibited alone, so co-targeting MET should preempt that escape route. SRC added as a third node to further block downstream signaling.

Critic agent trace:

Step 1 — Target overlap check (query BKG for shared downstream nodes) Retrieves pathway edges for all three targets. Finding: EGFR → RAS/MAPK, EGFR → PI3K/AKT, EGFR → SRC → STAT3. MET → RAS/MAPK, MET → PI3K/AKT, MET → SRC → STAT3.

Flag raised: **SRC sits downstream of both EGFR and MET, not parallel to them.** This isn't three orthogonal vulnerabilities — it's two upstream receptors converging on a shared node, plus that same shared node targeted a third time. The redundancy is real: inhibiting EGFR and MET already substantially suppresses SRC pathway flux. Adding a SRC inhibitor on top is closer to "hit the same pathway three times" than "hit three independent dependencies."

Step 2 — Does this redundancy actually matter, or is it reinforcing? This is where the Critic has to be careful not to over-correct — redundant hits on a convergence node can still be pharmacologically useful (deeper pathway suppression, higher bar for a bypass mutation to restore signaling). Checks the BKG for resistance-mutation edges: are there documented SRC-independent bypass routes from EGFR/MET (e.g., through STAT3 direct activation, or through PIK3CA mutation bypassing all three)? Finding: yes — PIK3CA activating mutations and STAT3 gain-of-function are both documented escape routes that don't require flux through SRC. So the triplet's redundancy doesn't buy as much resistance-durability as the original rationale implied; a fourth, truly parallel node (PI3K or STAT3) would do more for durability than the third redundant hit on the EGFR/MET/SRC axis.

Step 3 — Literature and trial history check Retrieves any prior trial or preclinical data for EGFR+MET combinations and SRC-inhibitor combinations in TNBC specifically. Finding: EGFR+MET dual inhibition has preclinical support in EGFR-amplified basal-like models; standalone SRC inhibitors (e.g., dasatinib) have underwhelmed in unselected TNBC trials, which is consistent with SRC being a redundant rather than independently rate-limiting node in this population.

Step 4 — Toxicity stacking check (Safety agent handoff) Three-way kinase inhibition raises overlapping toxicity concern: EGFR inhibitors and SRC inhibitors both carry GI toxicity liability; MET inhibitors add

hepatotoxicity risk. Three-drug overlap increases cumulative liver/GI burden without a corresponding three-fold gain in orthogonal efficacy, per Step 1-2 findings.

Verdict passed to Chair agent:

- **Mechanism validity: conditional pass, with a specific caveat** — not rejected, but re-scored down from "three independent hits" to "two upstream receptors plus one redundant convergence node."
- **Recommended modification:** consider swapping SRC for a node downstream of the documented bypass routes found in Step 2 — a PI3K or STAT3 inhibitor would close the escape routes that EGFR+MET+SRC leaves open, and would be a genuinely orthogonal third hit rather than a reinforcing one.
- **Selectivity/therapeutic index:** flagged as lower than the pairwise EGFR+MET combination alone, given added hepatotoxicity risk without proportional resistance-durability gain.
- **This gets written back to the BKG** as a new annotated edge: EGFR+MET+SRC — mechanistic-redundancy-flag — SRC downstream of both, so the next time any agent proposes a triplet built around a shared convergence node, that pattern is already flagged rather than re-discovered from scratch.

The thing this trace is meant to demonstrate: a Critic agent that just checked "is this plausible" would have passed this triplet — the rationale sounds coherent on its face. The value comes from forcing it to trace actual pathway topology and ask whether the third component is truly independent or just reinforcing, which is a distinction that's very easy for an LLM reasoning in prose (without graph grounding) to miss entirely.

I'll take that as "keep going" — starting with the clean-pass trace since it's the direct contrast to what we just saw, then I'll move to GNN code after.

Input from Hypothesis Generator:

Proposed triplet: FGFR1 inhibitor + JAK2 inhibitor + mTOR inhibitor, for basal-like/mesenchymal TNBC with FGFR1 amplification and active JAK/STAT signaling. Rationale: three parallel pathways (FGFR1 → RAS/MAPK, JAK2 → STAT3, PI3K/AKT → mTOR) hit simultaneously to prevent any single bypass route from restoring proliferative signaling.

Critic agent trace:

Step 1 — Target overlap check Retrieves pathway edges: FGFR1 → RAS/MAPK and FGFR1 → PI3K/AKT/mTOR (some overlap here, noted). JAK2 → STAT3, largely independent of the other two at the receptor level. mTOR sits downstream of PI3K/AKT, which does receive some input from FGFR1, but

mTOR is also the convergence point for nutrient/energy sensing pathways (independent of any of the three amplified receptors).

Finding: there is a partial overlap (FGFR1 feeds into the PI3K/mTOR axis to some degree), but unlike the EGFR/MET/SRC case, JAK2→STAT3 is a genuinely separate, non-convergent branch, and mTOR integrates inputs beyond just FGFR1 signaling. This reads as **two largely independent axes (FGFR1/MAPK and JAK2/STAT3) plus a shared downstream integration point (mTOR) that isn't simply redundant with either.**

Step 2 — Does each node close a distinct bypass route? Checks BKG for documented resistance mechanisms to FGFR1 monotherapy: STAT3 reactivation and PI3K/AKT reactivation are both known escape routes. Checks resistance to JAK2 monotherapy: FGFR1 upregulation and mTOR pathway reactivation are documented escapes. Finding: **each drug in this triplet closes a documented escape route for at least one of the other two** — this is the pattern you want. FGFR1 inhibition's main escape (STAT3/JAK reactivation) is covered by the JAK2 inhibitor; JAK2 inhibition's main escape (mTOR reactivation) is covered by the mTOR inhibitor; and mTOR's role as a general survival node is covered by removing upstream drive from both receptors.

Step 3 — Literature and trial history check FGFR1+JAK2 combination has preclinical rationale in FGFR1-amplified basal-like models with JAK/STAT co-activation. mTOR inhibitor addition to receptor-tyrosine-kinase-inhibitor backbones has trial precedent in other solid tumors as a resistance-preemption strategy, though direct TNBC triplet data is sparse (flagged as **novel — no direct precedent found**, which the Chair should note as higher uncertainty but not a rejection).

Step 4 — Toxicity stacking check (Safety agent handoff) mTOR inhibitors carry metabolic toxicity (hyperglycemia, stomatitis) that doesn't heavily overlap with FGFR1 inhibitor toxicity (hyperphosphatemia, retinal effects) or JAK2 inhibitor toxicity (myelosuppression). Toxicity profiles are largely non-overlapping — a genuinely favorable sign for a three-drug combination, versus the EGFR+MET+SRC case where GI/hepatic toxicity stacked.

Verdict passed to Chair agent:

- **Mechanism validity: clean pass.** Three genuinely complementary nodes, each closing a documented escape route for another.
- **Novelty flag:** no direct trial precedent for the three-way combination specifically — recommend prioritizing this as an early wet-lab candidate precisely because the in silico case is strong but empirical validation doesn't yet exist.
- **Selectivity/therapeutic index:** favorable, given non-overlapping toxicity profiles.
- **BKG update:** writes back FGFR1+JAK2+mTOR — validated-orthogonal-triplet — escape-route-complementary, which becomes a positive template pattern other hypotheses can be compared against.

The contrast is the point: same triplet-generation logic, same Critic checklist, but this one earns a clean pass because the graph traversal actually shows three non-redundant escape-route closures instead of one reinforced pathway. That's the difference between "sounds mechanistically sophisticated" and "is mechanistically sophisticated," and it's only checkable because the Critic is querying the graph structure rather than reasoning in prose.

Wet-Lab Experimental Spine of Hybrid-CORE

To make Hybrid-CORE scientifically credible, the computational engine must be anchored in a continuous wet-lab validation program. The laboratory component functions as the experimental spine of the closed loop. Every computational hypothesis generated by the system must eventually be tested against biological reality using cancer cell lines, patient-derived organoids, co-culture systems, or tissue-based models.

The central principle is:

Prediction is not evidence until it survives experimental testing.

Hybrid-CORE therefore follows a closed biological feedback cycle:

Hybrid-CORE proposes drug combinations

↓

Wet lab tests combinations in cell lines/organoids

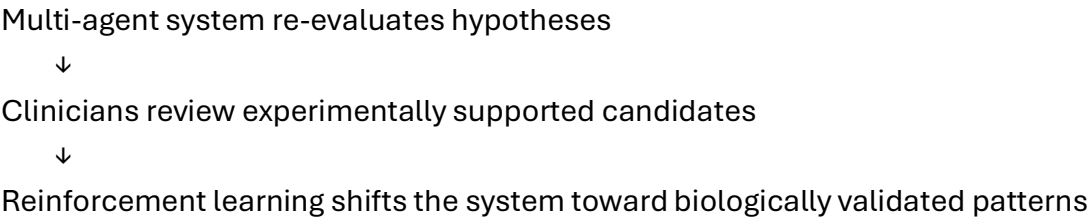
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Synergy, toxicity, viability, apoptosis, and pathway response are quantified

↓

Experimental results update KG++ and MM-SynergyNet

↓



Experimental Models

Hybrid-CORE can use multiple experimental systems:

Model	Purpose
Breast cancer cell lines	Fast initial screening of drug pairs/triplets
Patient-derived organoids	More realistic tumor architecture and heterogeneity
Normal breast epithelial cells	Tumor-selectivity and toxicity comparison
Co-culture models	Tumor-immune or tumor-stromal interaction testing
Patient-derived xenografts	Optional later-stage in vivo validation

Core Wet-Lab Readouts

Each experimental cycle should generate quantitative data that can be fed back into the model:

Readout	Biological Meaning	Model Feedback
Cell viability	Drug response strength	Updates synergy prediction
Bliss/Loewe/HSA/ZIP synergy	Combination interaction	Updates MM-SynergyNet labels
Apoptosis markers	Cell death mechanism	Updates mechanistic KG++
Phosphoproteomics	Pathway inhibition or compensation	Updates signaling edges
RNA-seq	Transcriptomic adaptation	Updates patient/tumor embeddings
Normal-cell viability	Toxicity/selectivity	Updates safety score
Organoid morphology	Structural response	Updates digital twin state

Feedback Into Hybrid-CORE Layers

The wet-lab data connects directly to the computational architecture:

Hybrid-CORE Layer	Wet-Lab Feedback
Multi-omics fusion	RNA-seq, proteomics, phosphoproteomics
KG++	New drug-target-pathway-phenotype relationships

MM-SynergyNet	Experimental synergy and selectivity labels
Multi-agent reasoning	Evidence Agent and Safety Agent update conclusions
RL optimization	Reward function updated from biological and clinician feedback

Why This Strengthens the Innovation

This wet-lab spine transforms Hybrid-CORE from an AI recommendation system into a genuine **AI-guided experimental discovery platform**. The system no longer depends only on published literature or retrospective datasets. Instead, it generates hypotheses, tests them, learns from failures, and improves over time.

This is the difference between:

Static AI ranking
and

Closed-loop AI-guided therapeutic discovery

The wet-lab component also makes the framework more attractive for publication, grant funding, pharmaceutical R&D, and translational oncology collaboration because it provides a concrete path from computational prediction to biological validation.

Safety Boundary

Hybrid-CORE should not be treated as a clinical prescribing system. Its outputs are experimentally supported research hypotheses that require oncologist review, pharmacist safety review, institutional approval, and clinical-trial validation before any patient use.

The **Hybrid-CORE Wet-Lab Validation** serves as the "experimental spine" of the framework, anchoring the computational engine in biological reality to ensure its predictions are scientifically credible. It transforms the system from a simple AI recommendation tool into a genuine **AI-guided experimental discovery platform** by continuously testing hypotheses against living biological models.

1. The Closed Biological Feedback Cycle

The validation follows a closed loop where every computational drug hypothesis—such as the **EGFR + MET + SRC triplet**—is tested in a laboratory setting. The resulting data is fed back into the system to refine the model's knowledge, allowing it to learn from both its successes and its failures.

2. Experimental Models and Their Purposes

The "spine" utilizes a hierarchy of biological systems to validate different aspects of a treatment's effectiveness:

- **Breast cancer cell lines:** Used for fast initial screening of drug pairs and triplets.
- **Patient-derived organoids:** Provide a more realistic look at tumor architecture and heterogeneity.
- **Normal breast epithelial cells:** Essential for comparing tumor-selectivity to ensure the therapy doesn't harm healthy tissue (toxicity comparison).
- **Co-culture models:** Test how the tumor interacts with immune or stromal cells.
- **Patient-derived xenografts (PDX):** Optional for late-stage *in vivo* (animal) validation.

3. Integrating Readouts into Hybrid-CORE Layers

Quantitative data from these experiments is used to update specific layers of the Hybrid-CORE architecture:

- **Multi-omics fusion:** Updated with new RNA-seq, proteomics, and phosphoproteomics data.
- **KG++ (Knowledge Graph):** Incorporates new drug–target–pathway–phenotype relationships discovered during testing.
- **MM-SynergyNet:** Refines its synergy and selectivity labels based on experimental scores (e.g., Bliss, Loewe, or HSA).
- **Multi-Agent Reasoning:** The **Evidence** and **Safety Agents** update their conclusions based on real-world viability and apoptosis markers.
- **RL Optimization:** The reinforcement learning reward function is updated using both biological feedback and clinician input.

4. Strategic Innovation and Safety

This approach strengthens the innovation by moving beyond reliance on published literature or retrospective datasets. It allows the system to generate and test its own novel hypotheses, making it highly attractive for pharmaceutical R&D and translational oncology.

However, the sources emphasize a **Safety Boundary:** Hybrid-CORE is a research and decision-support aid, not a clinical prescribing system. All outputs must be reviewed by oncologists, pharmacists, and institutional boards before any potential patient use.